

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12396721
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Smith; Adam Lee

Movable suturing apparatus and method

Abstract

Disclosed embodiments include apparatuses, systems, and methods for suturing a wound, incision, or other opening. In an illustrative embodiment, an apparatus includes a roller mechanism configured to counter-rotatably support two generally-parallel rollers configured to engage therebetween a shaft of a helically-shaped needle, the needle having a point at a leading end, drawing a filament from a trailing end, and having a first helical radius, the needle being revolvable around a first roller having a roller radius that is less than a helical radius of the needle to enable the needle to pierce a body and draw a filament through a body disposed adjacent the first roller. A motor drive is operably coupled with roller mechanism to convey rotational force to rotate the two generally-parallel rollers. A controller is operably coupled with the motor drive and configured to cause the needle to complete at least one revolution to form a single suture in the body.

Inventors:	Smith; Adam Lee (Palm Desert, CA)
Applicant:	Gyrus ACMI, Inc. (Westborough, MA)
Family ID:	1000008782587
Assignee:	Gyrus ACMI, Inc. (Westborough, MA)
Appl. No.:	17/215528
Filed:	March 29, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20210298749 A1	Sep. 30, 2021

Related U.S. Application Data

us-provisional-application US 63002995 20200331
us-provisional-application US 63002928 20200331

Publication Classification

Int. Cl.: **A61B17/06** (20060101); **A61B17/04** (20060101); **A61B17/062** (20060101); **A61B34/30** (20160101); A61B17/00 (20060101)

U.S. Cl.:

CPC **A61B17/06066** (20130101); **A61B17/04** (20130101); **A61B17/0482** (20130101); **A61B17/06128** (20130101); **A61B17/0625** (20130101); **A61B34/30** (20160201); A61B2017/00398 (20130101); A61B2017/00473 (20130101); A61B2017/0498 (20130101); A61B2017/06076 (20130101)

Field of Classification Search

CPC: A61B (17/06066); A61B (17/04); A61B (17/06128); A61B (34/30); A61B (2017/00398); A61B (2017/00473); A61B (2017/0498); A61B (2017/06076); A61B (17/0625); A61B (2017/00876); A61B (2017/047); A61B (17/0482); A61B (17/0491); A61B (17/11)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5545148	12/1995	Wurster	606/147	A61B 17/0469
6015416	12/1999	Stefanchik et al.	N/A	N/A
6613058	12/2002	Goldin	N/A	N/A
9962155	12/2017	Meade et al.	N/A	N/A
10736625	12/2019	Penn, IV	N/A	A61B 17/0469
2006/0212048	12/2005	Crainich	606/144	A61B 17/0469
2019/0290259	12/2018	Sikes et al.	N/A	N/A
2019/0343529	12/2018	Smith et al.	N/A	N/A
2019/0388087	12/2018	Almodovar	N/A	A61B 17/0469
2020/0375591	12/2019	Mohamed	N/A	A61B 17/0482
2021/0298735	12/2020	Smith	N/A	N/A
2022/0061836	12/2021	Parihar et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
113456138	12/2020	CN	N/A
102021107650	12/2020	DE	N/A
3108491	12/2020	FR	N/A
2590138	12/2020	GB	N/A
2595352	12/2020	GB	N/A
2013529982	12/2012	JP	N/A
2021159773	12/2020	JP	N/A

WO-2004066848	12/2003	WO	N/A
WO-2019246437	12/2018	WO	N/A

OTHER PUBLICATIONS

Sep. 10, 2021, International Search Report for Application No. GB2104561.2. cited by applicant
“U.S. Appl. No. 17/215,660, Non Final Office Action mailed Aug. 19, 2022”, 15 pgs. cited by applicant
“U.S. Appl. No. 17/215,660, Response filed Nov. 21, 2022 to Non Final Office Action mailed Aug. 19, 2022”, 14 pgs. cited by applicant
“French Application Serial No. 2103364, Office Action mailed Jul. 28, 2022”, with machine translation, 6 pgs. cited by applicant
“French Application Serial No. 2103364, Response filed Oct. 7, 2022 to Office Action mailed Jul. 28, 2022”, with English claims, 10 pgs. cited by applicant
“Japanese Application Serial No. 2021-057296, Notification of Reasons for Refusal mailed Mar. 20, 2023”, w/ English Translation, 14 pgs. cited by applicant
“U.S. Appl. No. 16/874,486, Response filed Jul. 28, 2023 to Advisory Action mailed Jun. 20, 2023”, 11 pgs. cited by applicant
“U.S. Appl. No. 17/215,660, Advisory Action mailed Sep. 7, 2023”, 3 pgs. cited by applicant
“U.S. Appl. No. 17/215,660, Final Office Action mailed May 31, 2023”, 18 pgs. cited by applicant
“U.S. Appl. No. 17/215,660, Non Final Office Action mailed Sep. 21, 2023”, 23 pgs. cited by applicant
“French Application Serial No. 2103364, Office Action mailed Oct. 5, 2023”, w/o English Translation, 1 pg. cited by applicant
“Japanese Application Serial No. 2021-057296, Notification of Reasons for Rejection mailed Sep. 19, 2023”, w/ English Translation, 8 pgs. cited by applicant
“Japanese Application Serial No. 2021-057296, Response filed Jun. 20, 2023 to Notification of Reasons for Refusal mailed Mar. 20, 2023”, w/ english claims. cited by applicant
U.S. Appl. No. 17/215,660, filed Mar. 29, 2021, Movable Suturing Apparatus and Method. cited by applicant

Primary Examiner: Woo; Julian W

Attorney, Agent or Firm: Schwegman Lundberg & Woessner, P.A.

Background/Summary

PRIORITY CLAIM (1) The present application claims the priority and benefit of U.S. Provisional Patent Application Ser. No. 63/002,928 and U.S. Provisional Patent Application Ser. No. 63/002,995, both filed Mar. 31, 2020 and entitled “MOVABLE SUTURING APPARATUS AND METHOD.”

FIELD

(1) The present disclosure relates to apparatuses, systems, and methods for suturing an object.

BACKGROUND

(2) The statements in this section merely provide background information related to the present disclosure and may not constitute prior art.

(3) Sutures are commonly used to close incisions or wounds. Sutures typically are formed by passing a curved needle through tissue at both sides of the incision or wound, with the needle drawing a filament from a trailing end. Once the curved needle completes a full revolution around

the incision or wound, the filament forms a suture to hold the tissue closed.

(4) Although the process itself is well known, the process of positioning the needle at a desired location and motivating the needle may not be simple. It may be difficult for a person suturing a wound or incision to reach the location and motivate the needle through the tissues, even when the suturing is performed by a skilled and/or experienced medical provider.

SUMMARY

(5) Disclosed embodiments include apparatuses, systems, and methods for suturing a wound, incision, or other opening.

(6) In an illustrative embodiment, an apparatus includes a roller mechanism configured to counter-rotatably support two generally-parallel rollers configured to engage therebetween a shaft of a helically-shaped needle, the needle having a point at a leading end, drawing a filament from a trailing end, and having a first helical radius, the needle being revolvable around a first roller having a roller radius that is less than a helical radius of the needle to enable the needle to pierce a body and draw a filament through a body disposed adjacent the first roller. A motor drive is operably coupled with roller mechanism to convey rotational force to rotate the two generally-parallel rollers. A controller is operably coupled with the motor drive and configured to cause the needle to complete at least one revolution to form a single suture in the body.

(7) In another embodiment, a system includes a helically-shaped needle having a helical radius, the needle having a point at a first end of a shaft and a filament joined to a second end. A roller mechanism is configured to counter-rotatably support two generally-parallel rollers configured to engage therebetween the shaft of the needle, the needle being revolvable around a first roller having a roller radius that is less than a helical radius of the needle to enable the first end of the needle to pierce a body and draw the filament through a body disposed adjacent the first roller. A motor drive includes a motor operably coupled with the roller mechanism to convey rotational force to rotate the two generally-parallel rollers and a controller operably coupled with the motor and configured to cause the needle to complete at least one revolution to form a single suture in the body.

(8) In a further embodiment, in an illustrative method, positioning a first roller of two generally-parallel rollers is positioned adjacent to a body, wherein the two generally-parallel rollers engage therebetween a helically-shaped needle, where the needle draws a filament from a trailing end, has a first helical radius that is greater than a radius of the first roller, and is revolvable around the first roller to enable the needle to pierce and extend through a body. An input is provided to a motor operably coupled with at least one of the two generally-parallel rollers, wherein the input causes the motor to rotate so as to cause the two-generally parallel rollers to rotate so as to cause the needle to complete at least one revolution to form a single suture.

(9) In an additional illustrative embodiment, an apparatus includes a helically-shaped needle having a first helical radius, the needle having a shaft and drawing a filament from a trailing end. Two generally-parallel rollers are configured to engage therebetween the shaft of the needle, the first roller having a roller radius that is less than the helical radius of the needle to enable the needle to revolve around the first roller to pierce a body disposed adjacent the first roller and to draw the filament through the body. A frame is configured to counter-rotatably support the rollers. The frame includes a support coupling configured to detachably receive a distal end of a support shaft extending from a drive mechanism and a drive coupling configured to detachably receive a distal end of a drive member extending from the drive mechanism, the drive member being configured to provide the rotational force to at least one of the rollers.

(10) In another illustrative embodiment, a drive mechanism includes a housing, a motor supported by the housing, a support shaft extending from the housing, and a drive member coupled with the motor and supported by the shaft where the drive member is configured to convey rotational force from the motor. A suture cartridge includes a helically-shaped needle having a first helical radius, the needle having a shaft and drawing a filament from a trailing end. Two generally-parallel rollers

are configured to engage therebetween the shaft of the needle, the first roller having a roller radius that is less than the helical radius of the needle to enable the needle to revolve around the first roller to pierce a body disposed adjacent the first roller and to draw the filament through the body. A frame is configured to counter-rotatably support the rollers. The frame includes a support coupling configured to detachably receive a distal end of the support shaft and a drive coupling configured to detachably receive a distal end of the drive member and to convey the rotational force to at least one of the rollers.

(11) In still another embodiment, an illustrative method includes engaging a suture cartridge with a motor drive. The suture cartridge supports a pair of generally-parallel rollers rotatably engaging a shaft of a helically-shaped needle therebetween, the needle being rotatable around a first roller and having a helical radius greater than that of a roller radius of the first roller to enable the needle to pierce and extend through a body disposed adjacent the first roller. The first roller of two generally-parallel rollers is positioned adjacent to the body. An input is provided to the motor drive, wherein the input causes the motor drive to rotate the two-generally parallel rollers to rotate so as to cause the needle to complete at least one revolution to form a single suture.

(12) Further features, advantages, and areas of applicability will become apparent from the description provided herein. It will be appreciated that the description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

DRAWINGS

(1) The drawings described herein are for illustration purposes only and are not intended to limit the scope of the present disclosure in any way. The components in the figures are not necessarily to scale, with emphasis instead being placed upon illustrating the principles of the disclosed embodiments. In the drawings:

(2) FIG. 1 is a perspective view of an embodiment of an apparatus for suturing an object;

(3) FIG. 2 is a perspective view of a roller mechanism of the apparatus of FIG. 1;

(4) FIG. 3 is a schematic block diagram of components of the apparatus of FIG. 1;

(5) FIGS. 4A and 4B are perspective views of the apparatus of FIG. 1 positioned to suture an object by a human hand and a robotic actuator, respectively;

(6) FIG. 5 is a side view of a helical needle usable with the apparatus of FIG. 1;

(7) FIG. 6 is a cross-sectional view of the needle of FIG. 5;

(8) FIG. 7 is a side view of the needle of FIG. 5 engaged by rollers of the apparatus of FIGS. 1 and 2;

(9) FIG. 8 is a top-down view of a portion of the needle of FIG. 5 received in a groove of one of the rollers of FIG. 7;

(10) FIG. 9 is a side view of the needle of FIG. 5 received in a of a roller having a flared groove;

(11) FIG. 10 is a schematic view of the rollers of the apparatus of FIG. 7 engaging the needle subject to a transverse force that compresses the needle;

(12) FIGS. 11-13 are schematic views of the rollers of the apparatus of FIG. 7 receiving the needle in a correcting recess;

(13) FIGS. 14-16 are side views of the needle being received by the correcting recess and guided into a groove;

(14) FIGS. 17 and 18 are schematic views of the grooved rollers of the apparatus of FIG. 7 employing other shapes of correcting recesses;

(15) FIGS. 19A and 19B are schematic, axial views of a needle potentially missing the correcting recess on the grooved roller;

- (16) FIGS. **20A** and **20B** are schematic, axial views of a needle having a reduced helical radius;
- (17) FIGS. **21A** and **21B** are schematic views of the needle of FIGS. **20A** and **20B** impacting on a surface of a grooved roller then being received in the correcting recess;
- (18) FIG. **22** is a perspective view of a shaft of the apparatus of FIG. **1** using a detachable suture cartridge;
- (19) FIG. **23** is a perspective view of a shaft of the apparatus of FIG. **1** engageable with alternative suture cartridges having differently-sized needles;
- (20) FIG. **24** is a top-down view of a supply tray that supports a motor drive of the apparatus of FIG. **1** and a number of interchangeable suture cartridges;
- (21) FIG. **25** is a side view of a portion of the supply tray of FIG. **24** that supports interchangeable suture cartridges;
- (22) FIG. **26A** is schematic diagram of a helical needle positioned to suture an opening;
- (23) FIG. **26B** is a schematic diagram of the helical needle of FIG. **26B** after it has been revolved to suture the opening;
- (24) FIG. **27** is a flow chart of an illustrative method of suturing a body with a helically-shaped needle motivated by two generally-parallel rollers; and
- (25) FIG. **28** is a flow chart of an illustrative method of engaging a suture cartridge with a motor drive to suture a body.

DETAILED DESCRIPTION

(26) The following description explains, by way of illustration only and not of limitation, various embodiments of apparatuses, systems, and methods for suturing an opening in a body, such as a wound or an incision. It will be noted that the first digit of three-digit reference numbers and the first two digits of four-digit reference numbers correspond to the figure number in which the element first appears.

(27) By way of a non-limiting introduction and overview, an apparatus and system of the present disclosure employ a motor drive to motivate two generally-parallel rollers that engage opposing faces of a shaft of a helical needle. A first of the two rollers is disposed adjacent to an opening in a body, such as a wound or an incision. The helical needle and the first roller have radii that are sized to enable the helical needle to revolve around the first roller and through tissues on opposing sides of the opening. A trailing end of the helical needle draws a filament. When the motor drive causes the rollers to revolve the needle through a full revolution, a suture is formed to close the opening. In various embodiments, the motor drive has a controller that, upon the pressing of a button or receipt of another discrete input, causes a motor to form a single suture. Thus, by positioning the rollers and the needle at the location where the suture is desired and providing a single input, a suture may be formed without an operator or actuator having to be moved to manually draw the needle through the body to form the suture.

(28) In various embodiments, the motor drive may be detachably engaged to a suture cartridge that supports the rollers and the needle. As a result, different cartridges may be detachably and disposably engaged by a motor drive to form sutures on different parts of a body or on different bodies. The motor drive may be reused, while suture cartridge including the rollers and needle may be discarded. Further, alternative suture cartridges may support differently-sized needles to support formation of differently-sized sutures for different applications.

(29) Now that an overview has been given, details of various embodiments will be explained by way of non-limiting examples given by way of illustration only and not of limitation.

(30) Referring additionally to FIG. **1**, in an apparatus **100** includes a motor drive **120** coupled with a roller mechanism **150** that motivates a helical needle **180** that draws a filament **190** to form a suture (not shown in FIG. **1**). The motor drive **120** includes a housing **122** that supports and/or contains other components of the motor drive **120**. The housing **122** includes an interface (e.g., suture control **124**, also referenced as control **124**) that facilitates the housing **122** being gripped by a human operator or supported by a robotic arm or other mechanical actuator (not shown in FIG.

1). The interface (e.g., suture control **124**) may be shaped or textured to aid in the operator or actuator being able to securely hold the housing **122**.

(31) The housing **122** may support one or more controls **124** and **126** that control operation of the motor drive **120**. In various embodiments, and as further described below, a suture control **124** may cause components of the motor drive **120** to revolve the needle **180** through one revolution to form a single suture. A second control **126** may be configured to stop and/or reverse the revolution of the needle **180** as desired. Electrical inputs **128** may be included to allow provision of power from an external source (not shown) to the motor and other components (not shown in FIG. 1). The electrical inputs **128** also may be used to provide input signals instead of or in addition to the controls **124** and **126**. For example, the use of such inputs **128** may allow a human operator to trigger the formation of a suture by activating a foot pedal (not shown) or other external control. In addition, if the apparatus **100** is employed by a robotic arm or other actuator, the inputs **128** may allow the actuator to control the apparatus **100** without having to physically engage the controls **124** and **126**.

(32) In various embodiments, a shaft **130** extends from the housing **122** to support the roller mechanism **150**. The shaft **130** may be sized and shaped to allow the roller mechanism **150** to be conveniently positioned away from the housing **122** to form sutures without the housing **122** and/or the operator's hand or other supporting body having to be placed close to the opening to be sutured. Distancing the housing **122** from the roller mechanism **150** thus may afford the operator greater visibility of the opening to be sutured and provide additional freedom of movement than if the housing **122** had to be positioned immediately adjacent the opening to be sutured. The shaft **130** may mechanically support the roller mechanism **150** and house a drive member (not shown in FIG. 1) that provides rotational force to the roller mechanism **150**, as further described below.

(33) Referring additionally to FIG. 2, the roller mechanism **150** may include a frame **154** that supports the rollers **156** and **158** that motivate the needle **180** which, in turn, draws the filament **190** that forms the suture. The frame **154**, as further described below, may be detachably coupled with the shaft **130**. A proximal end **152** of the frame **150** may engage a distal end **132** of the shaft **130**. The previously-mentioned drive coupling (not shown) may extend across and/or through the junction of the distal end **132** of the shaft **130** and the proximal end **152** of the frame to provide rotational force to the rollers **156** and **158**.

(34) The first roller **156** may be positioned adjacent to the opening in the body (not shown in FIGS. 1 and 2) to be sutured with the helical needle **180** being positioned to orbit around the first roller **156** to pierce the body and draw the filament **190** therethrough. A shaft of the needle **180** is engaged on opposing sides by the first roller **156** and the second roller, and the rollers **156** and **158** are counter-rotated to cause the needle **180** to revolve eccentrically around the first roller **156**. As further described below, the second roller **158** may include grooves **159** to partially receive the needle **180** to positionally secure the needle **180** as it is revolved to form a suture. An open end **160** of the frame **150** enables the filament **190** to be drawn by the needle **180** without the filament **190** having to be directed around or potentially being obstructed by a frame at the open end **160**.

(35) Referring additionally to FIG. 3, in various embodiments the motor drive **110** includes components that support the functions previously described. As previously described, the motor drive **110** includes the suture control **124**, the second control **126**, and the electrical inputs **128**. In such embodiments the motor drive **110** also may include a motor **310**, a controller **320**, a power supply **330**, and a drive member **340**.

(36) In various embodiments, the motor **310** is any type of electrical motor that provides a selectively-activated source of rotational force that is transmitted to the roller mechanism **150**. The motor **310** is mechanically coupled to a drive member **340**, such as a drive shaft or drive cable. As further described below, the drive member **340** is operably couplable with a drive coupling **350** of the roller mechanism **150**. The drive member **340**, which may be a drive shaft or drive cable that is mechanically coupled with at least one of the rollers **156** or **158**, receives the rotational force from

the motor **310** via the drive member **340** and the drive coupling **350** to rotate at least one of the rollers **156** and **158** to motivate the needle **180** to form sutures, as further described below.

(37) In various embodiments the motor **310** is coupled with the controller **320** which, in various embodiments, is coupled with the suture control **124**, the second control **126**, the electrical inputs **128**, and a power supply **330**. In various embodiments, the controller **320** is a logic circuit or microprocessor configurable, upon receipt of a control input received from the suture control **124**, the second control **126**, and/or the electrical inputs **128**, directs operation of the motor **310**. As previously described, when a user engages the suture control **124** (or a similar instruction is received via the electrical inputs **128**), the controller **320** causes the motor **310** to rotate to cause the roller mechanism **150** to revolve the needle **180** (FIGS. **1** and **2**) to complete a revolution to form a suture. The controller **320** receives power from the power supply **330** contained within the motor drive **110** or from an external power supply via the electrical inputs **128** and selectively provides electrical power to the motor **310** to cause the motor **310** to provide generate rotational force. As also previously described, when a user engages the second control **126** (or a similar instruction is received via the electrical inputs **128**), the controller **320** causes the motor **310** to stop or reverse, when desired.

(38) Referring additionally to FIGS. **4A** and **4B**, the apparatus **100** may be shaped to facilitate ease of operation. In various embodiments, the shaft **130** may include an angled portion **430** to enable the housing **122** to be positioned at a first displacement **432** from a body **410** to be sutured while the roller mechanism **150** is disposed at a location **412** where the suture is to be placed. The angled portion **430** allows for the first displacement **432** provides space for a hand of a human operator **404** (FIG. **4A**) or an actuator of an electromechanical device **452** (FIG. **4B**) supporting the apparatus **100** to hold the apparatus **100** in place without abutting the body **410**.

(39) Similarly, the shaft **130** may be configured to permit a second displacement **434** between the housing **120** and the roller mechanism **150** so that the motor drive **120** does not need to be placed immediately adjacent the location **412**. Allowing for the second displacement **434** may permit the roller mechanism **150** to be inserted at a location where the motor drive **120** may not fit because of other structures (not shown) or so as not to prevent the motor drive **120** from blocking a view of the location **412**.

(40) Referring again to FIG. **4A**, it will be appreciated that, with a user's hand **402** supporting the apparatus **100**, the user may hold the apparatus **100** with one hand at the location **412** and may initiate formation of a suture by engaging the suture control **124** with a human operator **404** (via a thumb or other finger, depending on placement of the suture control **124** and the hand). Referring again to FIG. **4B**, with the actuator **452** supporting the apparatus **100**, the actuator **452** may position the apparatus at the location **412** and may initiate formation of a suture by sending a control signal to the electrical input **128** via a signal coupling **454**. In either case, a suture may be formed at the location **412** without the effort or the space required for a person to use one or both of his hands at the location **412** to push and pull a needle through the body to form a suture.

(41) Referring additionally to FIG. **5**, the helical needle **180** has an axis **501** about which it revolves when motivated by the counter-rotating rollers **156** and **158** (FIGS. **1** and **2**). As shown below in FIGS. **1** and **2**, the axis **501** is not positioned coaxially with axes of either of the rollers **156** and **158**. Instead, the axis **501** is parallel with the axes of the rollers **156** and **158**, thereby enabling the needle **180** to be revolved by the rollers **156** and **158** while rotating about its own axis **501** and to engage the body **410** (FIGS. **4A** and **4B**) on a side of the needle **180** opposite the side of the needle **180** where it is engaged between the rollers **156** and **158**. In various embodiments, a shaft **510** of the needle **180** may have a rectangular or square cross-section, as described further below. The needle **180** has a pointed leading end **512** and draws the filament **190** from a trailing end **514**. In various embodiments, the filament **190** is coupled to the trailing end **514**, such as by the trailing end **514** being crimped around an end of the filament **190**.

(42) Referring additionally to FIG. **6**, in a section of the shaft **510** of the needle **180** taken along

axis 6-6 of FIG. 5, the shaft 510 of the needle 180 has opposing flat surfaces 602 and 604 on opposing sides of a generally rectangular or square cross-section 600. The flat surfaces 602 and 604 have a width $w_{sub.s}$. The opposing flat surfaces 602 and 604 are engageable on one side within the groove 159 of the roller 158 (FIGS. 1 and 2) and, on the other side, by a surface of the roller 156.

(43) Referring additionally to FIG. 7, the shaft 510 of the needle 180 is received within the groove 159 of the roller 158. In various embodiments, the groove 159 has a rectangular cross-section. The groove 159 is bounded by a generally flat bottom 712 that is bounded by generally flat sides 716 that each extend to an opening 714 at a grooved surface 720 of the roller 158. The roller 156 has a flat surface 757. The shaft 510 of the needle 180 is thus engaged between the groove 159 of the roller 158 at one side and the flat surface 757 of the roller 156 on an opposite side. The rollers 156 and 158 have open distal ends 756 and 758 at the open end 160 of the apparatus 100 (FIGS. 1 and 2), respectively. The open distal ends 756 and 758 of the rollers 156 and 158, respectively, allow the filament 190 to be drawn by the trailing end 514 (not shown in FIG. 7) of the needle 180 without the filament 190 being routed through or around a distal mount or other similar obstruction.

(44) The rollers 156 and 158 counter-rotate to revolve the needle 180, with the roller 156 rotating in a direction 766 and the roller 158 counter-rotating in an opposite direction 768. With the needle 180 disposed to revolve eccentrically about the roller 156, the needle 180 revolves in the same direction 756 as the roller 156. The rollers 156 and 158 may be mechanically interconnected, by gears or similar rotatable linkages (not shown in FIG. 7) to cause the rollers 156 and 158 to counter-rotate. Alternatively, because the rollers 156 and 158 are frictionally engaged with opposite sides of the shaft 510 of the needle 180, rotation of one of the rollers 156 or 158 would cause the opposite roller to counter rotate. In various embodiments, the leading end 512 of the needle 180 is disposed to advance in a direction 701 away from the open end 160 of the apparatus 100 as the rollers 156 and 158 are counter-rotated.

(45) Continuing to refer to FIG. 7, it will be appreciated that a pitch of the groove ϕ is ratiometrically-matched to a pitch of the needle θ according to a ratio of a helical radius R (FIG. 5) of the needle 180 to a radius r of the roller 158. As a result, the needle 180 will align with the groove 159 as the roller 158 is rotated along with the roller 156 in motivating the needle 180.

(46) Referring additionally to FIG. 8, with the groove 159 of the roller 158 having a width W larger than the width $w_{sub.s}$ of the shaft 512 of the needle 180 fits within the groove 159 without the sides 716 of the groove 159 impinging upon the sides 810 of the shaft 512 of the needle 180. As a result, the widened width W of the groove 159 avoids undesirable effects, such as compression of the needle 180 along its axis 501. In various embodiments, using a groove width W that is 20 percent larger than the width $w_{sub.s}$ of the shaft 510 of the needle 180 suitably accommodates the shaft 510 of the needle 180 within the groove 159 without the sides 716 impinging upon the shaft 510 of the needle 180.

(47) Referring additionally to FIG. 9, various embodiments of a grooved roller 958 may include a groove 959 that has two flared sides 916 to accommodate the shaft 512 of the needle 180. An object of the groove 959 with flared sides is to accommodate the sides of the needle 180. A square or rectangular groove without tapered sides may interfere with the needle 180 and cause the needle 180 to be deformed as the needle 180 rotates between the rollers. This tapered clearance is required to allow the helical needle to pass between the rollers 156 and 158 (FIG. 7) without with interference from the grooves. Thus, as shown in FIG. 9, the groove 959 flares from a bottom 912 of the groove 959 to an opening 914 at a surface 920 of the roller 958. Thus, the groove 959 has a generally trapezoidal cross-section. The width $w_{sub.F}$ of the groove 959 at the surface 920 is used to accommodate sides 912 of the shaft 510 of the needle 180 as the shaft 510 of the needle 180 when the needle 180 is displaced or deformed.

(48) Referring to FIG. 10, a force 1002 on the needle 180 also may cause the leading end 512 of the needle 180 to miss the groove 159 as the leading end 512 exits a body being sutured (not shown

in FIG. 10) and revolves back toward the grooved roller 158. Instead of revolving into the groove 159, the leading end 512 of the needle 180 impacts upon a surface 1004 of the roller 158 outside of the groove 159. It will be appreciated that the force 1002 could act in either direction and thereby potentially cause the leading end 512 of the needle 180 to miss the groove 159 on either side.

(49) Referring to FIG. 11, a correcting recess 1110 may be situated at a forward end 1112 of the groove 159 of the grooved roller 158. The correcting recess 1110 may be bounded by flared, lateral surfaces 1130 and 1131 that, as described below, will act as a funnel or guide to direct the leading end 512 of the needle 180 back into the groove 159. The lateral sides 1130 and 1131 define a widened area to receive the leading end 512 of the needle 180. Thus, the force 1002 does not act on the needle 180 (which would cause the leading end 512 of the needle 180 to miss the groove 159 and impact upon the surface 1004 of the roller 158). Instead, the leading end 512 of the needle 180 is received within the correcting recess 1110. Then, as described below, the counter-rotation of the rollers 156 and 158 will cause the correcting recess 1110 to rotate relative to the leading end 512 of the needle 180 to guide the needle 180 back into the groove 159.

(50) Referring to FIG. 12, as the rollers 156 and 158 counter-rotate with the grooved roller 158 rotating in a direction 1220, the correcting recess 1110 is rotated toward the opposing roller 156. The movement of the correcting recess 1110 results in the lateral surface 1130 of the correcting recess 1110 impinging upon the leading end 512 of the needle 180. The lateral surface 1130 moves the leading end 512 of the needle 180 in a direction 1240 to guide the leading end 512 toward the forward end 1112 of the groove 159.

(51) Referring to FIG. 13, with further rotation of the grooved roller 158 in the direction 1220, the leading end 512 of the needle is guided into the forward end 1112 of the groove 159. The correcting recess (not shown in FIG. 13) will repeatedly revolve into position to receive the leading end 512 of the needle 180 on a subsequent revolution of the needle 180 after forming each suture.

(52) Still referring to FIG. 13, in various embodiments, the groove 159 may have a reduced depth d.sub.1 at or near the forward end 1112 of the groove 159 as compared to a regular depth d.sub.2 of a remaining portion 1350 of the groove 159. The reduced depth d.sub.1 at the forward end 1112 may result in a reduced clearance between the forward end 1112 of the groove 159 and the opposing roller 156. Such reduced clearance could result in the opposing roller 156 applying higher pressure against the needle 180 at the forward end 1112 of the groove 159. The reduced depth d.sub.1 (and the resultant higher pressure) enables the rollers 156 and 158 to securely grip the needle 180 as the needle 180 begins its next revolution after having been redirected into the groove 159 by the correcting recess 1110.

(53) Referring to FIGS. 14-16, the leading end 512 of the needle 180 is directed by the lateral surface 1130 of the correcting recess 1110 into the forward end 1112 of the groove 159 of the grooved roller 158. Referring to FIG. 14, the leading end 512 of the needle 180 is received into the correcting recess 1110 between the lateral surfaces 1130 and 1131 that define the sides of the correcting recess 1110. As previously described with reference to FIG. 10, the leading end 512 of the needle 180 would have landed outside of the groove 159, the sides of which are represented by dotted lines in FIGS. 14-16.

(54) Referring to FIG. 15, as the grooved roller 158 is rotated as described with reference to FIGS. 12 and 13, the lateral surface 1130 of the correcting recess 1110 engages the leading end 512 of the needle 180. It will be appreciated that, as the grooved roller 158 is rotated as described with reference to FIG. 13, the correcting recess 1110 narrows between the lateral surfaces 1130 and 1131. Referring to FIG. 16, as the grooved roller 158 is further rotated as described with reference to FIG. 13, the lateral surfaces 1130 and 1131 of the correcting recess 1110 (not shown in FIG. 16) merge into the forward end 1112 of the groove 159. Thus, the correcting recess 1110 receives the leading end 512 of the needle 180 at a location outside of the groove 159 and directs the leading end 512 of the needle 180 into the groove 159.

(55) Referring to FIGS. 17 and 18, grooved rollers may include correcting recesses in other shapes

than that described with reference to FIGS. 11-16. Referring to FIG. 17, for example, the grooved roller **158** may include a correcting recess **1710** that is only widened on one side. For example, it may be anticipated that the only forces acting on the needle **180** will deflect the needle **180** in a direction **1702**. Thus, the correcting recess **1710** may include one lateral surface **1731** that extends straight from a forward end **1712** of the groove **159** and one flared lateral surface **1730** to accommodate deflection of the leading end **512** of the needle **180** in the direction **1702**. The single flared lateral surface **1730** should accommodate any anticipated deflection of the leading end **512** of the needle **180**.

(56) It will be appreciated that the lateral surfaces of the correcting recess are not limited to straight, flared shapes. Referring to FIG. 18, for example, a correcting recess **1810** may include one or more curved or other non-straight surfaces **1830** and **1831** to engage the leading end **512** of the needle **180**. The curved lateral surfaces **1830** and **1831** also may serve to impinge upon the leading end **512** of the needle **180** to guide the leading end **512** into the forward end **1812** of the groove **159**. The lateral surfaces of the correcting recess are not limited to any particular geometric shape.

(57) Referring to FIGS. 19A and 19B, based on forces that may affect movement of the helical needle **180**, such as resistance encountered by the needle **180** or slippage of the needle **180** between rollers **156** and **158**, may cause the leading end **512** of the needle **180** to miss the correcting recess **1110** when the leading end **512** rotates toward the correcting recess **1110**.

(58) Referring to FIG. 19A, a portion of the needle **180** is depicted as the needle **180** revolves around the roller **156**. It will be appreciated that the axial view of the needle **180** in FIGS. 19A and 19B, as well as the axial views of the needle **180** in FIGS. 20A and 20B, show a projection of the needle **180** that describe less than a full circular arc about their axes. In practice, however, the needle **180** actually describes more than a full circular arc around its axes to enable the rollers **156** and **158** to engage the shaft of the needle **180** as it fully revolves around its axis in forming sutures.

(59) The needle **180** describes more than a full circular arc around its axis, in part, to account for the narrowed, sharpened leading end **512** of the needles **180**, which may be too narrow to be engaged by the rollers **156** and **158**. Similarly, the trailing end **514** (FIG. 5) of the needle **180** may be narrowed or tapered to engage the filament it draws that is used to form the sutures (not shown in FIGS. 29A-20B) may be too narrow to be engaged by the rollers **156** and **158**. Accordingly, the needle **180** describes more than a full circular arc around its axis to allow for ends that may not be engageable by the rollers **156** and **158**. In any case, the needle **180** in FIGS. 19A-20B is represented as not describing a full circular arc around its axis to be able to show a position of the leading end **512** of the needle **180** without the position of the leading end **512** potentially being obscured by an overlapping portion of the needle **180**.

(60) Continuing to refer to FIG. 19A, as the needle **180** revolves in a direction **1901** as a result of the counter-rotation of the rollers **156** and **158**, the leading end **512** of the needle **180** approaches the correcting recess **1110** formed in the surface of the grooved roller **158**. Referring to FIG. 19B, when some forces as described have impeded movement of the needle **180**, the grooved roller **158** may have rotated such that the correcting recess **1110** has rotated such that the leading end **512** of the needle **180** is not received by the correcting recess **1110**. Generally speaking, a radius R of the helical needle **180** will be a whole number multiple of the radius r of the grooved roller so that movement of the correcting recess **1110** is synchronized with the revolution of the needle **180** to receive the leading end **512** of the needle **180**. In other words, a ratio between the radius R (or the diameter) of the needle **180** and the radius r (or the diameter) of the roller **158** will be a whole number that is greater than or equal to two. However, forces acting on the needle **180** may impair the synchronization of the rotation of the needle **180** and the grooved roller **158**, thus indicating that the needle **180** have a radius R that may be less than a whole number multiple of the radius r of the roller **158** (or that the ratio of R to r should be less than two or another whole number ratio of R to r).

(61) Referring to FIGS. 20A and 20B, reducing the radius of the helical needle **180** may

compensate for forces that may impede rotation of the needle **180** to avoid synchronization problems with the grooved roller **158**. Referring to FIG. **20A**, the helical needle **2090** has a radius R' that is approximately five percent less than the radius R of needle **180** (represented in dashed line form in FIG. **20A** for comparison with needle **2090**). Thus, in contrast to the needle **180** having a radius R that is twice that of the radius r of the grooved roller **158**, the needle **2090** has a radius R' that is approximately $1.9 R$, or five percent less than R . Having a shortened radius R' , by contrast with the needle **180** of radius R , the leading end **2092** of the needle **2090** is closer to the grooved roller **158** at this point in the revolution of the needle **2090** than was the leading end **512** of the needle **180**. Referring to FIG. **20B**, as the needle **2090** revolves in a direction **2001**, the leading end **2092** of the needle **2090** is received in the correcting recess **1110**.

(62) Referring to FIGS. **21A** and **21B**, a leading end **2092** of the needle **2090** may reach a surface **2104** of the grooved roller **158** before the correcting recess **1110** has been rotated into position to receive the leading end **2092** of the needle **2090**. Because the needle **2090** has a radius R' that is less than a whole number multiple of the radius r of the grooved roller **158** as previously described, the needle **2090** may complete a revolution before the grooved roller with radius r , less than one-half that of the needle **2090** in this example, completes two rotations. Thus, it is possible that the leading end **2092** of the needle **2090** may reach the surface **2104** of the grooved roller **158** before the correcting recess **1110** is in position to receive the leading end **2092** of the needle **2090**.

Referring to FIG. **21A**, the leading end **2092** of the needle **2090** may thus impact on the surface **2104** of the grooved roller **158** outside of the correcting recess **1110**.

(63) Referring to FIG. **21B**, however, as the grooved roller **158** continues to revolve in a direction **2101** in counterrotation with the roller **156**, the correcting recess **1110** rotates under the leading end **2092** of the needle **2090**. Thus, even if the leading end **2092** of the needle **2090** with radius R' should be in position to engage the correcting recess **1110** before the correcting recess **1110** is in position to receive the leading end **2092**, the leading end **2092** will slide along the surface **2104** of the grooved roller **158** until it can slide into the correcting recess **1110**.

(64) Even if the leading end **2092** should reach a gap **2125** (FIG. **21A**) between the surface **2104** of the grooved roller **158** and the surface of the opposing roller **156** before the leading end **2092** is received into the correcting recess **1110**, the needle **2090** cannot slip into the gap **2125** between the rollers **156** and **158**. It will be recalled and appreciated that the helical needle **180** or **2090** is sized to fit between the rollers **156** and **158** only when received within the groove **159** of the grooved roller **159**. Accordingly, the needle **2090** is too wide to fit into the gap **2125**. Thus, the leading end **2092** of the needle **2090** will slide along the surface **2104** of the grooved roller **158** until the correcting recess **1110** rotates beneath the leading end **2092** of the needle **2090**. At that point, the leading end **2092** of the needle **2090** will be captured by the correcting recess **1110** and guided into the groove **159** as previously described with reference to FIGS. **11-18**.

(65) In various embodiments, the roller mechanism **150** may be selectively detachable from and attachable to the motor drive **120**. The roller mechanism **150** thus may be changed for sanitary reasons and/or, as further described below, to engage needles of different sizes. As previously described, a proximal end **152** of the frame **154** of the roller mechanism **150** engages a distal end **132** of the shaft **130**. Referring additionally to FIG. **22**, in various embodiments the roller mechanism **150** may thus be disengaged by detaching a juncture of the proximal end **152** of the roller mechanism **150** and the distal end **132** of the shaft **130**. The roller mechanism **150**, including the frame **154**, rollers **156** and **158**, and needle **180**, may be regarded as a detachable and replaceable suture cartridge **2200**. The distal end **132** of the shaft **130** thus may include a shaft coupling **2230** configured to engage a support coupling **2210** at the proximal end **152** of the frame **150**. The couplings **2230** and **2210** may include a tongue-in-groove arrangement or another structure to frictionally and/or mechanically engage the distal end **132** of the shaft **130** with the proximal end **152** of the frame **154**.

(66) Still referring to FIG. **22**, in various embodiments the shaft **130** may support one or more drive

members **2260** and **2270** to convey rotational force from the motor **310** of the motor drive **110** (FIG. 3). The drive members **2260** and **2270** may include rigid or semi-rigid shafts or flexible drive cables. The drive members **2260** and **2270** may be coupled to the motor **310** in different ways. In various embodiments, the drive members **2260** and **2270** may be connected to a gear box (not shown) such that the drive members **2260** and **2270** counter-rotate in response to rotation of the motor **310** in order to counter-rotate the rollers **156** and **158**, respectively. Other embodiments, may include only a single drive member **2260** or **2270** to convey rotational force from the motor **310** to one of the rollers **156** or **158**. As previously mentioned, the rollers **156** and **158** may be coupled by a geared mechanism (not shown) housed at or within the frame **154** so that one of the rollers **156** or **158** receiving the rotational force from a drive member **2260** or **2270** will cause the other roller to counter-rotate. As also described, when one of the rollers **156** or **158** engages a drive member **2260** or **2270**, frictional engagement of the rollers **156** and **158** with the needle **180** will cause the other of the rollers to counter-rotate. Any of these configurations may be used whether the roller mechanism **150** is fixably attached to the motor drive **110** or if the roller mechanism **150** is part of a suture cartridge **2200** detachable from the motor drive **110**.

(67) Particularly when the motor drive **110** is selectively detachable from and couplable with the suture cartridge **2200**, the one or more drive members **2260** and **2270** must be detachable from and attachable to the rollers **156** and **158**. In various embodiments, at least one of the rollers **156** and/or **158** includes a shaft **2256** and/or **2266**, respectively, extending from the rollers **156** and/or **158** through the frame **154**. The shafts **2256** and/or **2266** are engageable by the drive members **2260** and/or **2270**, respectively. For purposes of the rest of this description, it will be understood that there may be one drive member **2260** or **2270** or two drive members **2260** and **2270** to engage one shaft **2256** or **2266** or two shafts **2256** and **2266**, respectively.

(68) To facilitate transfer of rotational force from the drive members **2260** and/or **2270** to the shafts **2256** and/or **2266**, each of the shafts **2256** and/or **2266** may include a drive coupling **2257** and **2267**, respectively. The drive couplings **2257** and **2267** include faceted ends **2259** and/or **2269**. The faceted ends **2259** and/or **2269** may include shaped outer ends, such as hexagonal ends as shown in FIG. 22, or may include support geared surfaces or other ends suitable to receive rotational force via the drive members **2260** and/or **2270**. The drive members **2260** and/or **2270** may include faceted ends **2262** and **2272** to engage the faceted ends **2259** and **2269**, respectively, of the drive couplings **2266** and **2268**. In various embodiments, the faceted ends **2262** and **2272** of the drive members **2260** and **2270**, respectively, may include inner-facing sockets to engage outer-facing ends of the drive couplings **2257** and **2267**, respectively. Thus, when the motor drive **110** is engaged with the suture cartridge **2200**, with the distal end **132** of the shaft **130** engaging the proximal end **152** of the frame **150**, the faceted ends **2262** and/or **2272** of the drive members **2260** and/or **2270** engage the faceted ends **2259** and **2269** of the drive couplings **2257** and/or **2267**, respectively.

(69) As previously mentioned, when interchangeable suture cartridges **2200** are used, different cartridges may have differently-sized needles for different applications. For example, a suture cartridge having a needle with a smaller helical radius may be used to form a small suture to close small wounds or incisions. Alternatively, a suture cartridge having a needle with a larger helical radius may be used to form a larger suture to close larger wounds or incisions. the suture cartridges may have differently-sized rollers. To accommodate differently-sized needles, pitches of the grooves of the grooved rollers and pitches of the needle should be ratiometrically matched to ensure proper alignment of the needle with the grooves as the grooved roller is rotated. In addition, it may be desirable to use differently-sized rollers to support differently-sized needles. For example, to suture a small incision, a needle with a small helical radius may be desirable so that the suture closely matches edges of the wound or incision to be closed. Accordingly, at least the roller about which the needle revolves should have a smaller radius to accommodate the needle having a smaller helical radius. Correspondingly, when a larger needle is desired to form a suture to close a

larger wound or incision, larger rollers may be desired to provide a wider surface with which to frictionally engage the needle as the needle is revolved to form the suture.

(70) Referring to FIG. 23, suture cartridges **2310** and **2311** of two different sizes support different sizes of needles **2380** and **2381** having different helical radii R and R' , respectively. Each of the suture cartridges **2310** and **2311** includes the support coupling **2210** configured to engage the shaft coupling **2230** of the shaft **130** so that the suture cartridges **2310** and **2311** may be mechanically supported by the shaft **130**. Both suture cartridges **2310** and **2311** include frames **154** of the same size. However, it will be appreciated that the frames of the suture cartridges **2310** and **2311** need not be of the same size as long as the frames are equally capable of engaging the shaft **130**.

(71) Both suture cartridge **2310** and **2311** include the shaft **2268** having the faceted end **2269** to engage the faceted end **2272** of the drive member **2270** to provide rotational force to the rollers. In contrast to the suture cartridge **2200** of FIG. 1, the suture cartridges **2310** and **2311** are motivated by the single drive member **2270**, with gears between the rollers or frictional engagement with the needle causing both rollers to counter-rotate, as previously described. The shaft **2268** and the faceted end **2269** of both suture cartridges **2310** and **2311** are equivalent to support interchangeability of the suture cartridges **2310** and **2311** with the drive member **2270**.

(72) The suture cartridge **2310** has a grooved roller **2358** with a radius r and an opposing roller **2356** around which revolves a needle **2380** of helical radius R . The suture cartridge **2311** has a grooved roller **2359** with a radius r' and an opposing roller **2357** around which revolves a needle **2381** of helical radius R' . The helical radius R' is greater than the helical radius R such that the suture cartridge **2311** is suited to form a larger suture than the suture cartridge **2310**. In the case of both the suture cartridge **2310** and the suture cartridge **2311**, the pitches of the grooves in the grooved rollers **2358** and **2359** and the helical needles **2380** and **2381**, respectively, are ratiometrically matched. The diameters r and r' are selected to cause the needles **2380** and **2381**, respectively, to revolve to form a single suture when the motor drive (not shown in FIG. 23) is engaged to cause the needle **2380** or **2381** to revolve. In various embodiments, the rollers **2356** and **2357** about which the needles **2380** or **2381** revolve, respectively, may be of the same size or of different sizes, provided that the rollers **2356** and **2357** provide suitable frictional engagement with the needles **2380** and **2381** to motivate revolution of the needles **2380** and **2381**.

(73) Although suture cartridges **2310** and **2311** of only two sizes are shown, it will be appreciated that there could be suture cartridges of any number of sizes to form sutures of any size. In addition, although not shown in FIG. 22, each of the needles **2380** and **2381** would be joined with a filament to effect formation of sutures.

(74) Referring additionally to FIG. 24, a supply tray **2400** supports a number of interchangeable suture cartridges **2411-2418**. The supply tray **2400** may include brackets **2420** to frictionally engage sides of the suture cartridges to hold the suture cartridges **2411-2418** in place until they are engaged by the motor drive **110** and dislodged from the supply tray **2400** for use. As described further with reference to FIG. 24, in various embodiments, the supply tray **2400** supports the suture cartridges **2411-2418** with the support couplings and drive couplings (not shown in FIG. 24) presented so that the motor drive **110** may engage any of the suture cartridges **2411-2418** without a user having to handle or touch the supply cartridges **2411-2418**. The user not having to handle or touch the supply cartridges **2411-2418** may prevent the suture cartridges **2411-2418** from being soiled or infected with germs or other matter from the user's hands. To this end, the supply tray **2400** may be covered with a sanitary wrapping **2402** (represented by dashed lines) to seal the supply tray. Similarly, the motor drive **110** (which also may be held to the surface of the supply tray **2400** by brackets **2422**) may be presented sanitarily on the supply tray **2400**.

(75) The suture cartridges **2411-2418** supported by the supply tray **2400** may be of different sizes to enable the operator to select an appropriate suture cartridge or cartridges for a present application. For example, the suture cartridge **2410** includes a smaller needle **2431** (and, thus, potentially smaller rollers, as previously described) than the needle **2433** of the suture cartridge **2413**.

Similarly, the suture cartridge **2417** includes a larger needle **2437** than the needle **2433** of the suture cartridge **2413**. The operator thus may select from among the available suture cartridges **2411-2418** to choose the suture cartridge that will form a suture of the size appropriate to the present application.

(76) Referring additionally to FIG. **25**, the supply tray **2400** presents ends of the suture cartridges **2411-2413** with each of their support couplings **2210** and drive couplings **2266** and **2268** presented to be accessible by the motor drive **110**. As shown in FIG. **24**, the suture cartridges **2411-2413** are mechanically held to the supply tray **2400** with brackets **2420** holding each of the suture cartridges **2411-2413** therebetween. Alternatively or additionally, the supply tray **2400** may include a base **2501** that forcibly engages the frame **154** of each of the suture cartridges **2411-2413** to hold the suture cartridges **2411-2413** in place. For example, the base **2501** may be an adhesive pad or a magnet configured to engage the frame **154** of each of the suture cartridges **2411-2413** to hold the suture cartridges **2411-2413** in place until engaged by the motor drive **110** to remove it from the supply tray **2400**.

(77) Referring additionally to FIGS. **26A** and **26B**, the revolution of the needle **180** by the rollers (not shown in FIGS. **26A** and **26B**) forms a suture **2650** to close an opening **2601** in a body **2600**, such as a wound or incision. As shown in FIG. **26A**, the needle **180** is about to be revolved in the direction **2602** to pierce the body **2600** adjacent the opening **2601**. Because the needle **180** is helical in shape, revolution of the needle **180** causes the needle **180** to be drawn in the direction **2603**. The leading end **512** of the needle **180** pierces the body **2600**. Revolution of the needle **180** draws the shaft **510** of needle **180** through the body until the trailing end **514** drawing the filament **190** emerges from the body **400**. As shown in FIG. **26B**, after the needle **180** is revolved in the direction **2602** and, thus, moved in the direction **2601**, a loop of the filament **190** has formed a suture **2650** through the body **2600**.

(78) Referring additionally to FIG. **27**, in various embodiments an illustrative method **2700** is provided for suturing an object by revolving a helical needle as previously described. The method **2700** starts at a block **2705**. At a block **2710**, a first roller of two generally-parallel rollers is positioned adjacent to a body, wherein the two generally-parallel rollers engage therebetween a helically-shaped needle. The needle draws a filament from a trailing end and having a first helical radius that is greater than a radius of the first roller and the needle is revolvable around the first roller to enable the needle to pierce and extend through a body. At a block **2720**, an input is provided to a motor operably coupled with at least one of the two generally-parallel rollers. The input causes the motor to rotate so as to cause the two-generally parallel rollers to rotate so as to cause the needle to complete at least one revolution to form a single suture. The method **2700** ends at a block **2725**.

(79) Referring additionally to FIG. **28**, in various embodiments an illustrative method **2800** is provided for using an interchangeable suture cartridge to form a suture. The method **2800** starts at a block **2805**. At a block **2810**, a suture cartridge is engaged with a motor drive, the suture cartridge supporting a pair of generally-parallel rollers rotatably engaging a shaft of a helically-shaped needle therebetween. The needle is rotatable around a first roller and has a helical radius greater than that of a roller radius of the first roller to enable the needle to pierce and extend through a body disposed adjacent the first roller. At a block **2820**, the first roller of two generally-parallel rollers is positioned adjacent to the body. At a block **2830**, an input is provided to the motor drive, wherein the input causes the motor drive to rotate the two-generally parallel rollers to rotate so as to cause the needle to complete at least one revolution to form a single suture. The method **2800** ends at a block **2835**.

(80) It will be appreciated that the detailed description set forth above is merely illustrative in nature and variations that do not depart from the gist and/or spirit of the claimed subject matter are intended to be within the scope of the claims. Such variations are not to be regarded as a departure from the spirit and scope of the claimed subject matter.

Claims

1. An apparatus comprising: a roller mechanism configured to counter-rotatably support two generally-parallel rollers configured to engage therebetween a shaft of a helically-shaped needle, the needle having a point at a leading end, drawing a filament from a trailing end, and having a first helical radius, the needle being revolvable around a first non-grooved roller configured to rotate in a first direction, the first non-grooved roller having a first roller radius that is less than a helical radius of the needle to enable the needle to pierce a body and draw a filament through a body disposed adjacent the first non-grooved roller, the roller mechanism including a second grooved roller with a second roller radius larger than the first roller radius; a motor drive operably coupled with roller mechanism to convey rotational force to rotate the two generally-parallel rollers, wherein the first roller and a second roller of the two generally-parallel rollers rotate in opposite directions; and a controller including an input circuit electrically coupled to a momentary switch configured to receive user input activating a control function to operate the motor drive to cause the needle to complete at least one revolution to form a single suture in the body.
2. The apparatus of claim 1, wherein the motor drive supports the roller mechanism at an offset relative to the roller mechanism, wherein the roller mechanism may be positioned at a proximate distance to the body while a handle remains positioned at a distance greater than the proximate distance from the body.
3. The apparatus of claim 1, wherein: the first non-grooved roller includes a flat surface configured to rollably engage a first face of the shaft of the needle; and a second roller configured to rotate in a second direction opposite the first direction, the second roller includes a grooved surface defining a groove that is configured to at least partially receive an opposite face of the shaft of the needle.
4. The apparatus of claim 3, wherein a bottom of the groove has a bottom width at least as wide as the second face and bounded by sides extending to a groove opening at the grooved surface having a surface width that is wider than the second face.
5. The apparatus of claim 3, wherein a pitch of the groove is ratiometrically matched to a pitch of the needle according to a ratio of the helical radius of the needle to the second radius of the second roller.
6. The apparatus of claim 3, wherein the second roller further defines a correcting recess bounded by lateral surfaces at a forward end of the groove that tapers to a width of the groove at the forward end of the groove, and wherein when the leading end is received into the correcting recess responsive to counter-rotation of the first and second rollers and at least one of the lateral surfaces engages and guides the leading end into the forward end of the groove.
7. The apparatus of claim 1, wherein the roller mechanism includes an open distal end wherein ends of the two-generally parallel rollers are exposed, wherein with the trailing end of the needle faces the open distal end to draw the filament from the open distal end.
8. The apparatus of claim 1, wherein the roller mechanism is configured to receive the rotational force from the motor drive and impart the rotational force to at least one of the first roller and the second roller.
9. The apparatus of claim 1, wherein the controller is configured to receive a momentary input configured to cause the needle to complete at least one revolution to form a single suture in the body.
10. The apparatus of claim 1, wherein the controller is configured to receive a reverse input configured to cause the motor drive to revolve the needle in a reverse direction.
11. A system comprising: a helically-shaped needle having a helical radius, the needle having a point at a first end of a shaft and a filament joined to a second end; a roller mechanism configured to counter-rotatably support two generally-parallel rollers configured to engage therebetween the shaft of the needle, the needle being revolvable around a first roller upon rotation of the first roller

in a first direction and a second roller in a second direction, the first roller including a smooth surface and having a first roller radius that is less than the helical radius of the needle to enable the first end of the needle to pierce a body and draw the filament through a body disposed adjacent the first roller, and wherein the second roller includes a grooved surface engaging an outer surface of the helically-shaped needle and a second roller radius larger than the first roller radius; and a motor drive including: a motor operably coupled with roller mechanism to convey rotational force to rotate the two generally-parallel rollers; a controller operably coupled with the motor and configured to cause the needle to complete at least one revolution to form a single suture in the body; and an interface configured to enable the motor drive to be supported by one of a human user and a robotic actuator.

12. The system of claim 11, wherein the motor drive supports the roller mechanism at an offset relative to the roller mechanism, wherein the roller mechanism may be positioned at a proximate distance to the body while a handle remains positioned at a distance greater than the proximate distance from the body.

13. The system of claim 11, wherein a bottom of a groove in the grooved surface has a bottom width at least as wide as the second face and bounded by sides extending to a groove opening at the grooved surface having a surface width that is wider than the second face.

14. The system of claim 11, wherein a pitch of a groove in the grooved surface is ratiometrically matched to a pitch of the needle according to a ratio of the helical radius of the needle to the second radius of the second roller.

15. The apparatus of claim 11, wherein the second roller further defines a correcting recess bounded by lateral surfaces at a forward end of a groove that tapers to a width of the groove at the forward end of the groove, and wherein when the leading end is received into the correcting recess responsive to counter-rotation of the first and second rollers and at least one of the lateral surfaces engages and guides the leading end into the forward end of the groove.

16. The system of claim 11, wherein the roller mechanism includes an open distal end wherein ends of the two-generally parallel rollers are exposed, wherein with the trailing end of the needle faces the open distal end to draw the filament from the open distal end.

17. The system of claim 11, wherein the controller is configured to receive a momentary input configured to cause the needle to complete at least one revolution to form a single suture in the body.

18. The system of claim 11, wherein the controller is configured to receive a reverse input configured to cause the motor drive to revolve the needle in a reverse direction.

19. A method comprising: positioning a first non-grooved roller of two generally-parallel rollers adjacent to a body, wherein the two generally-parallel rollers engage therebetween a helically-shaped needle, the first non-grooved roller including a first radius and a second grooved roller of the two generally-parallel rollers including a second radius larger than the first radius, the needle drawing a filament from a trailing end and having a first helical radius that is greater than the first radius of the first non-grooved roller, wherein the needle is revolvable around the first non-grooved roller to enable the needle to pierce and extend through a body; and providing an input via an input control electrically coupled to a motor operably coupled with at least one of the two generally-parallel rollers, wherein the input causes the motor to rotate so as to cause the two-generally parallel rollers to each rotate in a different direction so as to cause the needle to complete at least one revolution to form a single suture.
